Sri Lanka Institute of Information Technology



SE2030 – Software Engineering
Year 2, Semester 1- 2026

Web-Based Event Planning System (Evenza)

Proposal Report

2026-Y2-S1-MLB-B10G1-02

IT Number	Name
IT25101514	Chathuranga B. G.
IT25102322	Wickramarathna W. L. S.
IT25103301	Keshavan M.
IT25100916	Suriyage O. W.
IT25102923	Gunasekara N. B.
IT25100220	Wickramasinghage C. P. W.

1. Introduction

1.1 Background and Problem Statement

In modern business operations, managing events manually through traditional tools—such as direct phone calls, emails, messaging platforms (e.g., WhatsApp), and spreadsheet tracking—often results in significant communication gaps, double-booking errors, delayed approvals, and scattered records.

The proposed Web-Based Event Planning System is a centralized software platform designed to streamline and automate the entire lifecycle of an event. The platform unifies event coordinators, venue owners, vendors, finance/administrative officers, inventory staff, customers, and guests into a single cohesive Web Application.

1.2 What is the project about?

The project is a Web-Based Event Planning System. It is a centralized online platform designed to replace manual, fragmented event management methods—such as phone calls, emails, WhatsApp messages, and spreadsheets—with a unified digital workspace.

manage the entire lifecycle of an event in one place—from initial creation, venue and vendor bookings, and task scheduling, inventory allocation and customer communication.

1.3 Objectives

- **Centralization**: Consolidate event planning, venue/vendor bookings, task scheduling, inventory allocation, and customer records into one centralized system.
- **Automation**: Streamline approval workflows, real-time availability checking, and instant scheduling notifications.
- **Error Reduction**: Eliminate double bookings, missing inventory, lost booking requests, and communication delays.
- **Improved Coordination**: Provide role-specific dashboards for smooth communication between event planners, customers, vendors, and management.

1.4 Target Users and Stakeholders

- **Administrator**: System overseer responsible for platform governance, user role management, system reporting, and platform maintenance.
- **Customer**: End-user who creates events, browses and books venues or vendors, tracks progress, and views financial details.

- **Event Manager/Coordinator**: Professional who verifies event plans, creates activity schedules, assigns staff, and coordinates execution.
- **Vendor/Service Provider**: External partner (caterer, decorator, photographer, sound operator) offering services, checking schedule availability, and managing bookings.
- **Inventory Staff**: Operational staff managing event assets (tables, chairs, audio equipment) and stock availability.
- **Guest/Attendee**: External participant who views event schedules, registers, and receives notifications.

1.5 Scope and Limitations

Scope: User authentication, role-based access control, event creation/modifications, vendor/venue search and booking, visual calendar/schedule management, inventory tracking, and customer management.

Limitations: Native offline mode (requires active internet access), multi-currency dynamic FX conversion, and hardware-level RFID physical inventory tag synchronization.

2. System Overview Diagram

The System Overview Diagram illustrates the core structural breakdown of the Web-Based Event Planning System, highlighting how individual specialized modules communicate through a unified central platform. At the foundation of the system is the Central Database, which stores all persistent data, including user credentials, event schedules, vendor listings, asset counts, and customer profiles. Operating above the database layer are the six primary core modules.

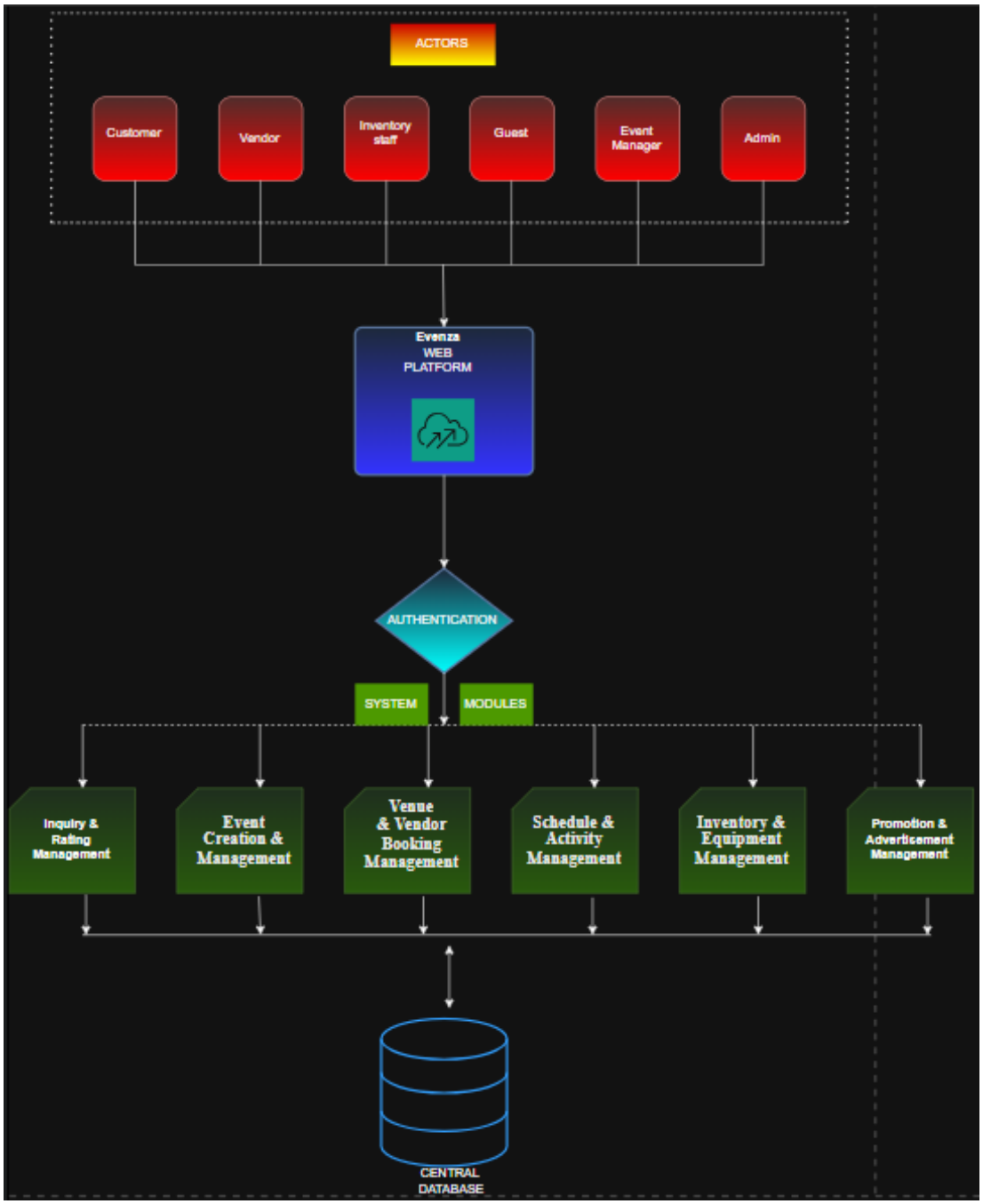


Figure 2.1 – System Architecture of the Web-Based Event Planning System

2.1 How the System Works

The Web-Based Event Planning System allows different users to access the application according to their assigned roles. After authentication, users interact with the appropriate system modules. Customers create events and submit booking requests. Event Managers coordinate schedules and approve activities. Vendors manage service bookings, while Inventory Staff manage equipment allocation. All modules exchange data through a centralized database, ensuring consistency and preventing double bookings.

3. Functional Requirements

FR1: Inquiry, Feedback & Rating Management

- The system allows guests and attendees to submit event-related inquiries and receive direct responses from event coordinators or vendors.
- The system shall allow verified attendees to submit post-event feedback and numerical ratings (e.g., 1–5 stars) for events and participating vendors.
- The system shall aggregate and display feedback metrics and performance rating summaries on vendor profiles and manager dashboards.

FR2: Event Management

- The system shall allow users to create, update, view, and cancel events.
- The system shall track event approval states (Draft, Pending Approval, Approved, Completed, Cancelled).

FR3: Venue & Vendor Booking

- The system shall display verified venues and vendors filtered by availability, service type, and price.
- The system shall allow customers to submit direct booking requests and prevent double-booking through real-time calendar locks.
- The system shall allow customers to make payments for confirmed venue and vendor bookings and record the payment status for each booking.

FR4: Schedule & Activity Management

- The system shall generate visual calendar views and timelines for scheduled events.
- The system shall allow event coordinators to assign specific tasks and schedules to operations staff.

FR5: Inventory Tracking

- The system shall record and update equipment stock levels (chairs, tables, AV systems) upon assignment to an event.
- The system shall send alerts to inventory managers when item stock falls below safe operational limits.

FR6: Promotion & Advertisement Management

- The system shall allow vendors and service providers to create, publish, and manage promotional offers, package discounts, and advertisement banners.
- The system shall display relevant vendor promotions and advertised packages to customers during event browsing and vendor selection workflows.
- The system shall track promotional campaign engagement (impressions, clicks, and inquiry conversions) and provide analytical summaries to vendors.

4. Non-Functional Requirements

- **Performance:** The platform shall support up to 500 concurrent active users with page response times under 2.0 seconds.
- **Security:** All user data, credentials, and transaction tokens must be transmitted over HTTPS with SSL/TLS encryption. Passwords must be hashed using bcrypt before storage.
- **Usability:** The Web interface shall be fully responsive across mobile devices, tablets, and desktop browsers, requiring minimal technical training.
- **Availability & Reliability:** The system shall maintain 99.5% operational uptime with daily automated database backups.
- **Scalability:** The backend architecture shall be horizontally scalable using modular microservices/REST APIs to accommodate future additions.

5. Major Stakeholders / Users

Stakeholder	Primary Responsibilities	System Value Derived
System Administrator	Manages system accounts, user permissions, audit logs, and reports.	Full platform control and system maintenance capability
Customer	Creates events, submits vendor/venue bookings, tracks timelines.	Single dashboard to organize events without scattered channels
Event Manager	Verifies event plans, coordinates timelines, assigns staff.	Automated scheduling, task tracking, and approval management.
Vendor / Service Provider	Lists services, manages availability, accepts/rejects bookings.	Direct channel to customers, automated booking management.
Inventory Staff	Manages physical stock allocation and returns.	Prevents equipment shortages and item losses
Guest / Attendee	Views event details, schedule times, and confirms attendance.	Convenient access to event information.

6. Major Functions

Each of the six major functions below is assigned to one group member, who is responsible for its design, implementation, and testing, and who will present it during the proposal presentation

Although each function is owned individually, the six modules are interdependent: venue and vendor booking depends on both user registration and event creation being complete, while schedule and inventory management rely on approved event details and stock records to accurately coordinate activities and allocate resources. The team will therefore integrate modules incrementally throughout each sprint, rather than leaving integration to the end of the project and will use shared Git branches with regular merges to keep every member's work compatible with the rest of the system.

6.1 Function 1: Inquiry, Feedback & Rating Management

- **Assigned Member**: Chathuranga B. G.
- **Target User**: Guest, Customer, Vendor, Event Manager
- **Detailed Description**: Enables guests, customers, and event attendees to submit event-related inquiries, deliver structured post-event feedback, and rate venues and vendors to enhance service quality and transparency across the platform.
- **Sub-functions**:

    1. Submit direct inquiries regarding event details or vendor services to event coordinators and vendors.
    2. Submit post-event reviews and numerical ratings (e.g., 1–5 stars) for completed events and participating vendors.
    3. View aggregated rating scores, attendee feedback, and review histories on vendor profiles.
    4. Respond to attendee inquiries and review comments.
    5. Generate and view performance feedback analytics and rating summaries on vendor and manager dashboards.

- **Expected Outcome**: Streamlined inquiry handling, transparent vendor performance tracking, and actionable feedback metrics for service improvement.

6.2 Function 2: Event Creation & Management

- **Assigned Member**: Suriyage O. W.
- **Target User**: Customer, Event Manager
- **Detailed Description**: Serves as the operational core allowing users to draft, publish, modify, and monitor event details throughout their lifecycle.
- **Sub-functions**:
    1. Create new events with date, venue, guest count, and requirements.
    2. View comprehensive event summary and status.
    3. Update/modify existing event parameters.
    4. Cancel/delete planned events.
    5. Track real-time progress of event approvals.
- **Expected Outcome**: Efficient creation and status management of events without manual paperwork.

6.3 Function 3: Venue & Vendor Booking Management

- **Assigned Member**: Gunasekara N. B.
- **Target User**: Customer, Vendor, Event Manager
- **Detailed Description**: Facilitates searching, availability verification, and formal booking requests between customers and service providers.
- **Sub-functions**:
    1. Search and filter available venues and service vendors.
    2. Perform real-time slot and availability checks.
    3. Submit booking requests directly to vendors.
    4. Accept, reject, or modify booking requests by vendors.
    5. Process event payments and maintain payment status records.
    6. Cancel/modify active vendor reservations.
- **Expected Outcome**: Streamlined vendor reservations that eliminate double bookings and miscommunications.

6.4 Function 4: Schedule & Activity Management

- **Assigned Member**: Wickramarathna W. L. S.
- **Target User**: Event Manager, Operational Staff, Guest
- **Detailed Description**: Organizes event timelines into structured tasks, calendar views, and milestone schedules.
- **Sub-functions**:
    1. Construct master event activity timelines.
    2. Assign specific tasks to operational staff members.
    3. Update schedule task execution status.
    4. Display visual multi-event calendar views.
    5. Automated notification and task deadline reminders.
- **Expected Outcome**: Clear operational visibility and milestone tracking for on-time event execution.

6.5 Function 5: Inventory & Equipment Management

- **Assigned Member**: Manirenkan K.
- **Target User**: Inventory Staff, Event Manager, Admin
- **Detailed Description**: Tracks physical asset allocation, equipment availability, and usage of lifecycles per event.
- **Sub-functions**:
    1. Add and categorize physical inventory items.
    2. View active inventory stock records and allocations.
    3. Update item quantities upon stock additions or damages.
    4. Allocate/reserve equipment for specific events.
    5. Record post-event equipment returns and condition reports.
- **Expected Outcome**: Complete accountability of event equipment preventing shortages and operational delays.

6.6 Function 6: Promotion & Advertisement Management

- **Assigned Member**: Wickramasinghage C. P. W.
- **Target User**: Vendor, Service Provider, Customer, Admin
- **Detailed Description**: Empowers vendors and service providers to create, publish, and manage promotional offers, package discounts, and advertisement banners to expand market reach and attract customers.
- **Sub-functions**:

    1. Create, publish, and update promotional offers, package discounts, and advertisement banners.
    2. Display relevant targeted vendor promotions and featured ad listings to customers during event browsing and selection.
    3. Manage active promotional campaign timelines, terms, and discount parameters.
    4. Track ad campaign performance metrics including impressions, clicks, and inquiry conversions.
    5. Archive or deactivate expired promotional offers and advertisement listings.

Expected Outcome: Expanded marketing visibility for vendors, targeted offers for customers, and measurable promotional performance analytics.

7. Minor Functions

- **User Authentication & Authorization (Login/Logout)**: Standard session handling ensuring secure system entry and termination.
- **Password Reset & Recovery**: Email-based recovery links or temporary verification codes allowing users to regain access.
- **Automated System Notifications**: Email/in-app alert mechanisms sending real-time status updates (e.g., booking approvals, schedule updates).
- **Basic Data Reporting**: Simple statistical outputs generating PDF/CSV summaries for event costs, booking history, and system usage.

8. System Limitations / Constraints

- **Platform Constraints**: The initial web application will require modern browser capabilities (Chrome, Firefox, Safari, Edge) with JavaScript enabled.
- **Network Dependence**: Real-time synchronization requires an active internet connection; no offline data caching will be supported in Version 1.0.
- **Third-Party API Limits**: Integration with external email and messaging gateways may be subject to third-party API rate limits and operational costs.
- **Scope Assumption**: System payments are recorded as payment status logs in V1.0; direct payment gateway handling (e.g., Stripe/Pay Here) is assumed as an optional future expansion.

9. Simple Timeline (Week 3 – Week 14)

The table below outlines the group's planned activities from proposal submission through to the final viva, aligned with the three formal evaluation checkpoints of the module:

Week Range	Key Expected Activities & Milestones
Week 3	Project Topic Finalization, Stakeholder Requirements Gathering, Proposal Submission & Presentation.
Week 4 - 5	Software Requirement Specification (SRS) Drafting, Use Case Diagrams, and System Architecture Design.
Week 6-7	Database Schema Design (ER Diagram), Interface Wireframing, and Tech Stack Setup.
Week 8 - 10	Core Backend API Development & Database Implementation (Functions 1 to 6).
Week 11 - 12	Frontend Interface Development, API Integration, and Unit/Integration Testing.
Week 13	User Acceptance Testing (UAT), System Bug Fixing, Performance Optimization, and Report Writing.
Week 14	Final Code Freeze, System Deployment, and Final Project Presentation / Submission.

10. Conclusion

The proposed **Web-Based Event Planning System** directly addresses the key operational challenges faced by event management organizations—including fragmented communications, manual tracking errors, double bookings, and resource misallocations. By consolidating management, bookings, timelines, inventory, and customer interactions into a unified platform, the system significantly improves operational efficiency, accuracy, and overall customer satisfaction.

With a clear, balanced division of responsibilities across six core modules, our team is well-structured to complete development successfully according to the 14-week project timeline